Faculty Evaluation System

Northwest Samar State University (NWSSU)

Technical guide: how the system works, how data flows, and how the code is arranged

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1. Overview and technology

The Faculty Evaluation System is a web application. Students rate their teachers, supervisors (department chairs/deans) rate the faculty in their departments, and administrators manage the records and produce the official CHED **CMO No. 19 s. 2025** reports (Annex C, Annex D and the Faculty Evaluation Report).

It has **three kinds of users**:

User	Logs in at	What they do
Administrator	/index.html (email + password)	Manages school years, students, teachers, subjects, departments, courses, question sets and the evaluation period; views results; prints reports.
Student	/Student-user/index.html (Student ID + password)	Evaluates each teacher of each enrolled subject using the SET form (Annex A, 15 questions).
Supervisor	/Student-user/index.html (Teacher ID + password)	A teacher record with facultyType = "supervisor". Evaluates faculty in the departments they supervise using the SEF form (Annex B).

Technology stack

- **Plain HTML, CSS and JavaScript.** No framework, no build step, no npm. Each page loads its scripts with <script> tags.
- **Firebase JS SDK 10.12 (compat version)** loaded from gstatic.com: *firebase-app*, *firestore*, *auth* (and *functions* on the admin page).
- **Cloud Firestore** is the database. **Firebase Authentication** handles sign-in. **Firebase Hosting** serves the site (firebase.json sets the repo root "." as the public folder).
- **SheetJS (xlsx 0.18.5)** on the admin page for Excel export / import.
- Firebase project ID: **studenteval-937f6** (in .firebaserc). An Android app (separate code, "AuthRepository.kt") uses the same project and the same login conventions.

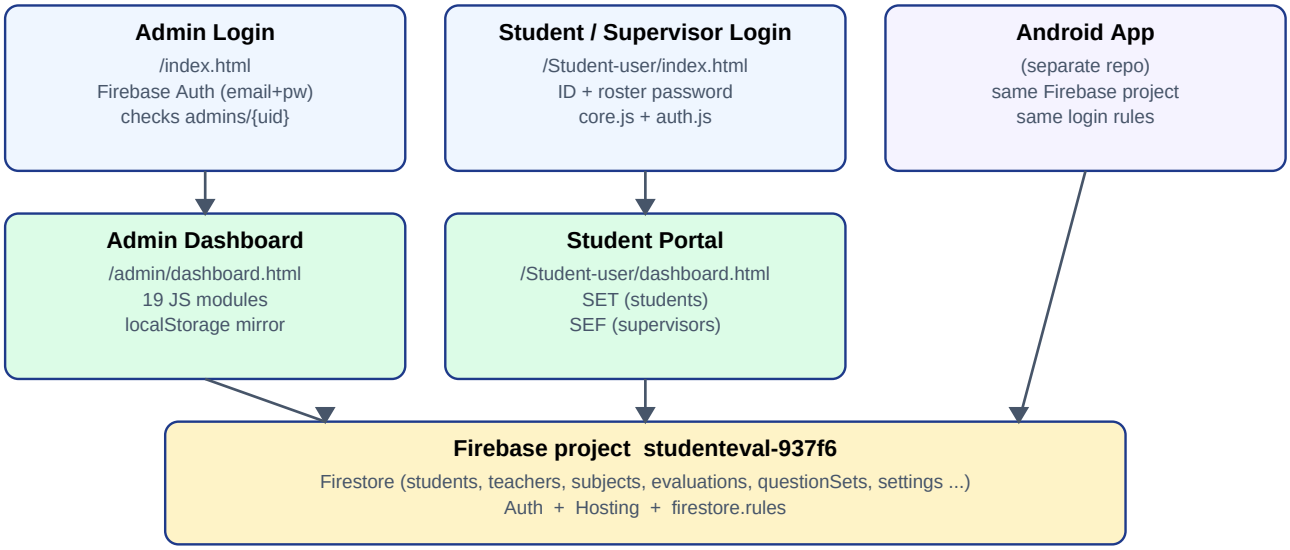
2. Folder structure and file arrangement

```
Faculty-Eval-System/
|-- index.html            Admin login page (entry point of the site)
|-- index.css             Styles for the admin login page
|-- 404.html              Not-found page for Firebase Hosting
|-- firebase.json         Hosting + Firestore rules config
|-- .firebaserc           Firebase project id (studenteval-937f6)
|-- firestore.rules       Database security rules
|-- shared/
|   |-- js/firebaseAPI.js Shared Firebase web config (window.firebaseConfig)
|   `-- css/styles.css    Shared styles
|-- icons/                Logos and department images (CCIS, CEA, COED ...)
|-- Student-user/         STUDENT + SUPERVISOR PORTAL
|   |-- index.html        Login page (ID + password, forced password gate)
|   |-- dashboard.html    The portal app (dashboard, evaluate, history ...)
|   |-- terms.html        Terms page
|   |-- css/user.css      Portal styles
|   `-- js/
|       |-- core.js       Firebase init, shared state, questions, helpers
|       |-- auth.js       Login, provisioning, password change, session restore
|       `-- app.js        Screens: dashboard, SET/SEF forms, history, profile
`-- admin/                ADMIN DASHBOARD
    |-- dashboard.html    Single-page admin app (all pages are <div>s)
    |-- admin-styles.css  Admin styles
    `-- js/
        |-- core/         admin-core, admin-scoring, admin-nav, admin-router
        |-- records/      students, teachers, subjects, bulk CSV, courses
        |-- reports/      reports, annex (C/D), fer, sef (+ audit log)
        `-- modules/      admindept, adminrate, adminEval,
                          adminfeatures, adminQuestions
```

How the arrangement works

- **Two separate apps share one Firebase project.** The admin app lives under */admin*; the student/supervisor app lives under */Student-user*. They never call each other directly; they meet only in Firestore.
- **Each app is split into a login page and a dashboard page.** The login page signs in and then redirects (location.replace) to the dashboard, so the Back button does not return to the login form.
- **The admin JS is grouped by job**: *core/* (storage, scoring, navigation, URL routing), *records/* (CRUD for people, subjects and courses), *reports/* (printed forms and analytics), *modules/* (departments, sync, term logic, features, questions). These files were split from one original 4,441-line admin.js.
- **The student JS is three layers**: core.js (data and helpers) -> auth.js (who you are) -> app.js (what you see). They must load in that order.

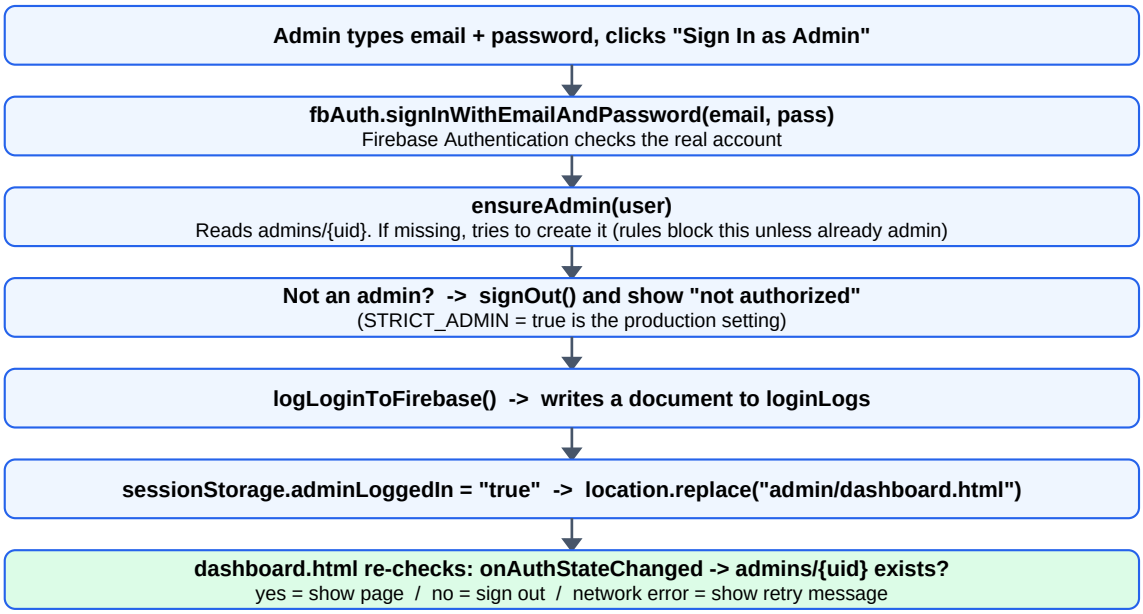
3. System architecture



Everything runs in the browser. There is no custom backend server: the browser talks directly to Firestore and Firebase Auth, and **firestore.rules** is what decides who may read or write each document. The admin dashboard additionally keeps a **copy of every collection in localStorage** so screens render instantly, and keeps that copy fresh with live listeners (section 9).

4. Admin login flow

File: **/index.html** (inline script). Function: **doAdminLogin()**.



- **First admin bootstrap:** create the document *admins/<your-uid>* once in the Firebase console. Rules forbid any client from writing to *admins*.
- On load, the page deletes old keys (adminCreds, adminPassword ...) left in localStorage by an earlier insecure version.
- If the admin is already signed in and returns to the login page, it bounces straight back to the dashboard.
- dashboard.html hides itself (visibility:hidden) until the admin check passes, and has an 8-second watchdog that explains the problem (usually opening the file as file:// instead of through a web server).

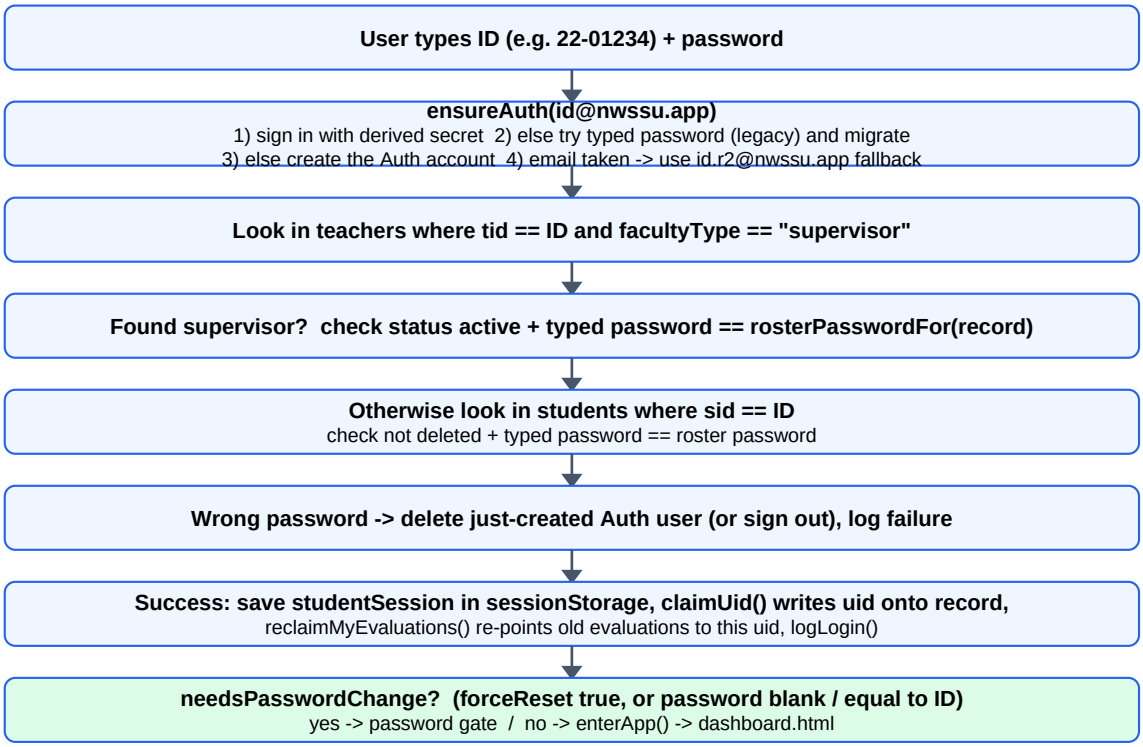
5. Student and supervisor login flow

Files: **Student-user/js/core.js** and **auth.js**. Function: **doLogin()**.

The key idea: there are two passwords.

Password	Where it lives	Who knows it
Roster password (the one the person types)	The *password* field on their students/{doc} or teachers/{doc} record. Blank = their own ID.	The person and the admin.
Firebase Auth secret (hidden)	Derived by authSecretFor(id) = "nwssu:<id>:v1". The Auth email is <id>@nwssu.app.	Nobody types it. The web and Android code compute it.

This lets the admin reset or change a password just by editing the roster field in Firestore, without needing admin access to Firebase Auth.



Default password = the person's ID. New and bulk-imported students get forceReset: true, so on first login they must choose a new password (min 6 characters, not their ID). Inactive students can still log in but see a yellow "inactive" banner.

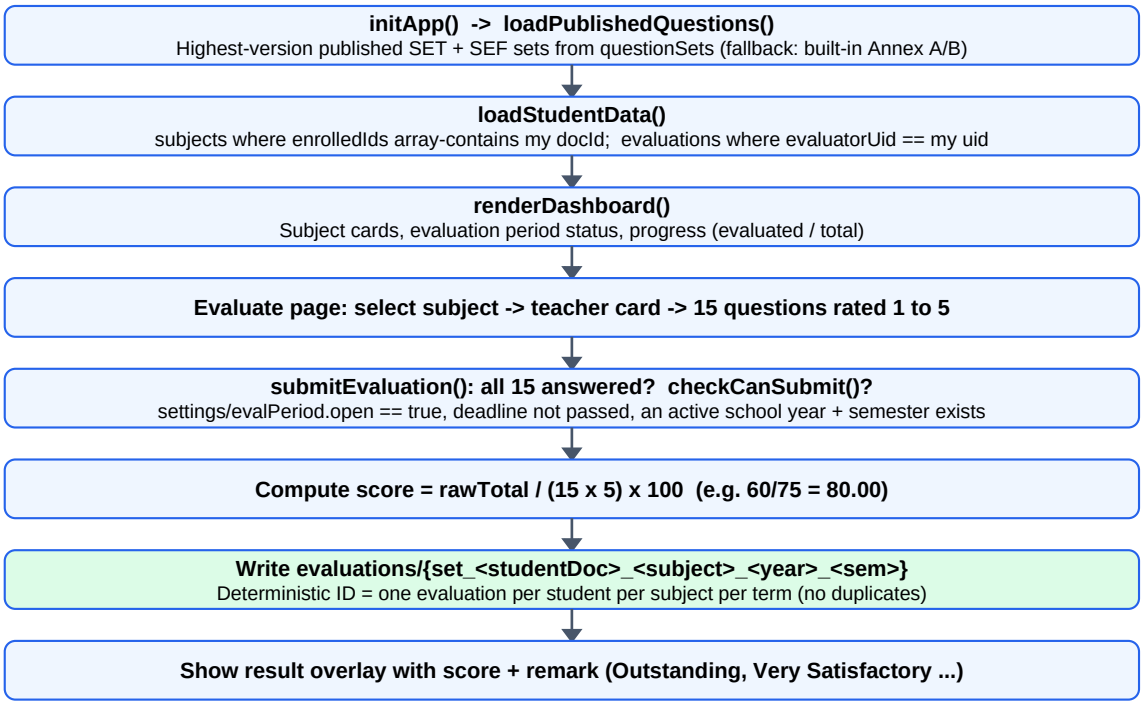
6. Session restore and the password gate

Both portal pages load auth.js. It detects which page it is on (IS_DASHBOARD = the #app element exists) and after *window.load* runs **restoreSession()**:

- **On the login page:** if already signed in with a valid record, go straight to the dashboard.
- **On the dashboard:** no studentSession, or no Firebase user, -> back to login with a clear reason message.
- It re-reads the roster record with _**readRosterDoc()**, which returns *ok / missing / denied / unreachable* (2 retries with timeouts). **Unreachable** (bad wifi) keeps the session and shows a Retry screen instead of logging the person out.
- A hop counter (_hop) stops endless login <-> dashboard redirect loops after 2 hops.
- **submitNewPassword()** (forced gate) and **changePassword()** (Settings) update only *password* and *forceReset* on the person's own record, which is exactly what the security rules allow.

7. Student evaluation flow (SET, Annex A)

File: **Student-user/js/app.js**.



Evaluation record fields

```
{ id, studentId, subjectId, teacherId, evaluatorType: "student", evaluatorUid,
  ratings: { q1: 5, q2: 4, ... q15: 5 }, rawScore, maxRaw: 75, totalScore,
  questionSetId, comment, schoolYear: "2025-2026", semester: "1st Semester",
  timestamp }
```

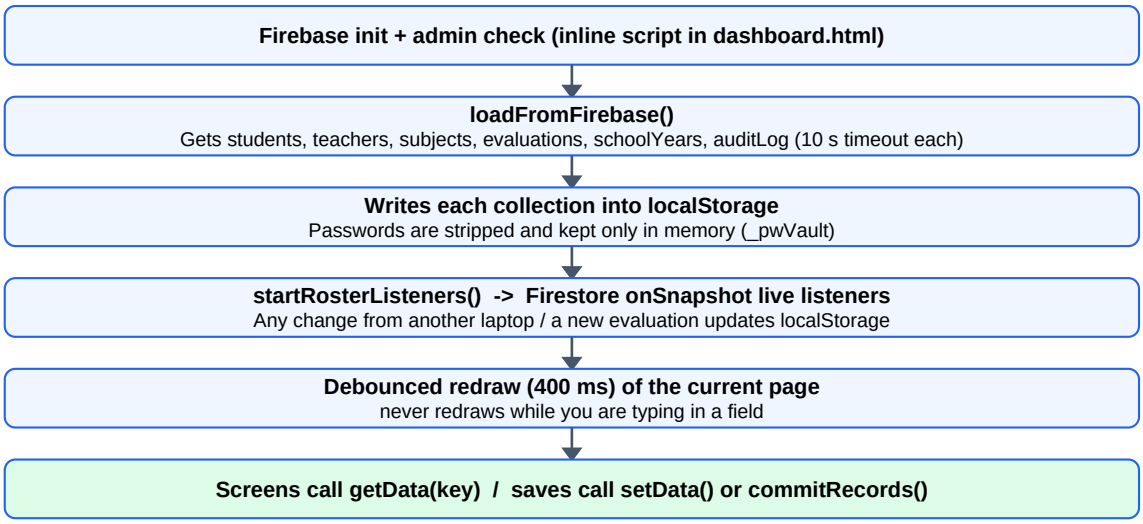
Other student screens: **My Evaluations** (history), **My Feedback** (own ratings + comments; opens a read-only view that shows the question wording of the version that was actually answered), **Profile** and **Settings** (change password). Pages are switched by **showPage(id)**, which toggles the *.active* class on *#page-<id>* divs.

8. Supervisor evaluation flow (SEF, Annex B)

- A supervisor is a **teacher record** with *facultyType: "supervisor"* and *supervisedDepts: [{dept, role}]*.
- initApp() sees userType "supervisor" and calls **renderSupervisorHome()**: **loadSupervisorData()** lists regular faculty in every department they supervise, plus their past SEF submissions.
- **startSef(teacherId)** reuses the Evaluate page with the Annex B questions (items 2, 4 and 9 differ from Annex A).
- **submitSef()** runs the same checks, then writes *evaluations/sef_<supervisorDoc>_<teacher>_<year>_<sem>* with evaluatorType "supervisor" and supervisorId. Submitting again in the same term **updates** the rating instead of duplicating it.

9. Admin dashboard: boot, data layer and sync

The admin dashboard is a single HTML page (**admin/dashboard.html**, ~1,800 lines) with every page as a hidden <div id="page-..."">. Its most important design decision: **screens read from localStorage, not directly from Firestore.**



Data layer functions (admin/js/core/admin-core.js)

Function	What it does
getData(key)	Reads JSON from localStorage. For evaluations, hides ones whose evaluator was deleted (countedEvaluations). For students/teachers, puts passwords back from the in-memory vault.
setData(key, value)	Saves to localStorage and calls syncCollectionToFirestore() to push the collection (used for subjects, schoolYears, settings, questionSets ...). Blocked for students/teachers until passwords have loaded.
commitRecords / writeRecordToCloud	Used for students and teachers. Writes ONLY the changed fields of ONLY the changed record, so two admins on two laptops do not overwrite each other.
commitRecordsBatch	Same, in a Firestore batch (bulk CSV import).
addAudit(action, detail)	Adds an entry to the auditLog (append-only; only new entries are sent).
resetLoginPassword()	Admin reset: sets the roster password (default = ID) and forceReset false.
syncCollectionToFirestore (adminrate.js)	Final version of the sync. Maps keys to collections (departments and customCourses become one doc per department; evalPeriod goes to settings/evalPeriod). Also deletes docs removed locally, but only if the last load was complete (_firestoreLoadClean).

Why the password vault? Earlier versions saved every student's password in localStorage in plain text. Now passwords stay in memory for the session only; closing the tab forgets them. Firestore still holds the real value.

URL routing

admin-router.js wraps showPage() so the current page is kept in the URL hash (e.g. *dashboard.html#reports*). Reload and the Back button therefore return to the same page.

10. Script load order (critical)

The admin scripts are plain global scripts. **When two files define the same function, the one loaded LAST wins.** The order in dashboard.html is therefore part of the program:

#	File	Role
1	shared/js/firebaseAPI.js + inline script	Config, Firebase init, admin check, loadFromFirebase, live listeners
2	core/admin-core.js	getData / setData, password vault, audit, toast, sorting, skeletons

#	File	Role
3	core/admin-scoring.js	All rating maths (weighted SET, SEF, final rating, remarks)
4	core/admin-nav.js	showPage, sidebar, logout, dashboard, school years
5-7	records/admin-students, teachers, subjects	CRUD for people and subjects
8-12	reports/admin-reports, records/admin-bulk, reports/admin-annex, admin-fer, admin-sef	Eval control, analytics, CSV import, Annex C/D, FER, SEF entry, audit page
13	records/admin-courses.js	Courses by department, supervisor migration, page bootstrap
14-18	modules/admindept, adminrate, adminEval, adminfeatures, adminQuestions	Departments; final sync + subject teachers; term logic, exemptions, backup; drill-downs and feedback; question versions
19	core/admin-router.js	MUST be last: wraps the final showPage

Known overrides: *saveSubject, renderSubjects, populateTeacherSelect* in admin-subjects.js are dead code (adminrate.js redefines them). *renderStudents* in admin-students.js is replaced by the one in adminfeatures.js. *syncCollectionToFirestore* in dashboard.html is replaced by the one in adminrate.js. Edit the LAST definition.

11. Admin pages and modules

Page (hash)	Purpose	Main file(s)
#dashboard	Counts and overview	admin-nav.js
#schoolYear	Add school years / semesters, set the ACTIVE semester	admin-nav.js
#students	Add / edit / deactivate / delete students; ID format YY-NNNNN; CSV bulk upload; reset password	admin-students.js, admin-bulk.js, adminfeatures.js
#teachers	Faculty and supervisors (tabs), multi-department supervisor assignments	admin-teachers.js
#subjects	Subjects, curriculum slot (courses + year level), enrolment, one or more teachers per subject (optionally per section)	admin-subjects.js, adminrate.js
#evalControl	Open/close the evaluation period, set the deadline, submission tracking per department / faculty	admin-reports.js
#reports	Faculty and supervisor result tables, filters, term picker, Annex C / D, FER, PDF/Excel export	admin-reports.js, admin-annex.js, admin-fer.js, adminEval.js
#supervisor	Admin-entered SEF (Annex B)	admin-sef.js
#auditLog	Read-only history of admin actions	admin-sef.js
#dept / #deptManage	Department drill-downs and department management (no hard-coded departments)	admindept.js, adminfeatures.js
#feedback	Student comments	adminfeatures.js
Questions modal	Edit and publish new versions of the SET / SEF questions	adminQuestions.js
Backup / Restore	Download / import a JSON backup of all collections	adminEval.js

12. Question sets (versioned instrument)

Evaluations save answers by question ID ({q1: 4, q2: 5 ...}). If a question's wording were edited in place, old answers would silently be reported against the new wording. So **adminQuestions.js** uses versions:

- A **published** set is never edited. Editing creates a **draft**; publishing it creates version n+1 and marks version n **retired**.
- Stored in the **questionSets** collection: { id, instrument: "SET" | "SEF", version, status, questions: [{id, sec, text}], publishedAt, retiredAt, note }.
- The portal and the Android app load the highest published version; every evaluation stores **questionSetId**, so reports show the wording that was actually answered.

- Default questions are CMO 19 Annex A / B **verbatim** (section 4.2 forbids changing the indicators). Sections: A Management of Teaching and Learning (q1-q6), B Content Knowledge, Pedagogy and Technology (q7-q11), C Commitment and Transparency (q12-q15).

13. Scoring and ratings (CMO 19)

All rating maths is in **admin/js/core/admin-scoring.js**, so every report uses the same numbers.

Step 1 - one evaluation

```
score = (sum of 15 ratings) / (15 x 5) x 100          e.g. 63 / 75 x 100 = 84.00
```

Step 2 - weighted SET per faculty: computeWeightedSET() (CMO 8.3)

- Take the faculty's subjects, **excluding** Overload subjects (CMO 4.3) and Lab-school subjects (CMO 8.5).
- For each subject keep only the students that teacher actually teaches (a teacher assigned to sections is rated only by those sections).
- Group the student evaluations by year/section (e.g. "BSIT 4A"). For each section: average score x number of respondents = weighted score.
- Sections or classes with no responses are listed but do not pull the rating down. Exempted students (CMO 8.2) are removed from the roll.
- SET_WEIGHT_BY = "respondents": the count is the students who actually submitted (not the class size).

```
overall SET = sum(avg_section x respondents) / sum(respondents)   (capped at 100)
```

Step 3 - SEF: getSEFForTeacher() (CMO 9.3)

The simple average of all supervisor (SEF) evaluations for that teacher in the term.

Step 4 - final rating: calculateFinalRating() (institutional, not CMO)

```
final = SET x 0.60 + SEF x 0.40      (only when the teacher has BOTH)
otherwise: "Awaiting SEF" or "No evaluations yet"
```

Remarks bands (same in core.js and admin-scoring.js)

Score	Remark	Colour
90 - 100	Outstanding	green
75 - 89.99	Very Satisfactory	blue
60 - 74.99	Satisfactory	amber
50 - 59.99	Fair	orange
below 50	Unsatisfactory	red

- Evaluations from deleted students/supervisors are hidden from every count (isEvaluationHidden). Inactive people still count.
- Printed annexes report SET and SEF separately, as CMO 19 requires; the 60/40 composite is only on internal screens.

14. Term scoping and reports

- **Active term:** the semester marked active under School Year. Every evaluation is stamped with schoolYear + semester.
- **adminEval.js**: _evalMatchesTerm() decides if an evaluation belongs to a term. setReportTerm() lets the admin browse a past term or "ALL". Untagged legacy records appear only in the active term.
- **Evaluation period** (settings/evalPeriod: open, deadline): students can only submit while open. **checkAndAutoFinalize()** runs on a timer; when the deadline is reached it closes the period, writes an audit

entry and runs generateFinalReports() (saved to finalReports).

- **Annex C** (buildAnnexCContent): per-class / per-section weighted SET table; clicking the count shows who has and has not submitted.
- **Annex D** (buildAnnexDContent): summary with SET and SEF, plus the FEDAF development plan (developmentPlans).
- **FER** (generateFERReport, admin-fer.js): institutional Faculty Evaluation Report and trend charts.
- **Signatories** (Prepared / Reviewed / Approved by) are saved once and filled into every printed report; fields remain editable before printing.

15. Firestore data model

Collection	Contents (main fields)	Written by
admins/{uid}	role, email - marks a Firebase user as an administrator	Firebase console only
students	sid (YY-NNNNN), name, course, year, section, dept, password, status, forceReset, deleted, uid	Admin; student may change own password / uid
teachers	tid, name, dept, facultyType ("regular"/"supervisor"), supervisedDepts[{dept, role}], password, status, deleted, uid	Admin; supervisor may change own password / uid
subjects	code, name, courses[], yearLevel, teacherId, teachers[{id, sections[]}], enrolledIds[], loadType, isLabSchool	Admin
evaluations	See section 7. IDs: set_... (student) / sef_... (supervisor)	Students / supervisors (own); admin
questionSets	Versioned SET / SEF instruments	Admin
schoolYears	year, semesters[{label, active}]	Admin
settings	evalPeriod {open, deadline} and other settings docs	Admin
departments / courses	One doc per department code; courses/{DEPT} = {dept, courses[]}	Admin
finalReports, developmentPlans, exemptions	Generated results, FEDAF plans, CMO 8.2 exemptions	Admin
auditLog	Admin actions (append-only)	Admin
loginLogs	username, success, timestamp, userAgent, loginType	Anyone (create only)

Relationships

```
student.id  --in-->  subject.enrolledIds[]
subject.teachers[].id / teacherId  -->  teacher.id
evaluation.studentId -> student.id     evaluation.subjectId -> subject.id
evaluation.teacherId -> teacher.id     evaluation.supervisorId -> teacher.id (SEF)
evaluation.questionSetId -> questionSets.id
evaluation.evaluatorUid  -> Firebase Auth uid (= student.uid / teacher.uid)
```

16. Security rules (firestore.rules)

- **isAdmin()** = signed in AND admins/{uid} exists. Admins can write all management collections.
- Reference data (questionSets, subjects, schoolYears, settings, departments, courses, teachers) is readable by any signed-in user.
- **students**: a student can read only their own record (matched by the ID in their login email). They can update only *password, forceReset, uid*.
- **teachers**: supervisors can likewise update only those three fields on their own record (uses get('tid', '') so missing fields do not break the rule).
- **evaluations**: create only with evaluatorUid == your own uid; read/update your own; delete admin-only. ownsEvaluation() lets a re-created account re-claim old evaluations by changing only evaluatorUid.
- **auditLog** is append-only; **loginLogs** accepts creates from anyone; everything not listed is denied.

17. Important notes and maintenance tips

- **Run through a web server.** Opening the HTML files directly (file://) breaks the login session between pages. Use Firebase Hosting, VS Code Live Server or *python -m http.server*.
- **Deploy:** *firebase deploy* (hosting + rules). Rules changes take effect only after deploying firestore.rules.
- **The Firebase config is repeated in 3 places**: shared/js/firebaseAPI.js, index.html (inline) and Student-user/js/core.js (plus a fallback in admin/dashboard.html). Change all of them if the project changes. The apiKey is a public identifier, not a secret.
- **STRICT_ADMIN** appears in both index.html and admin/dashboard.html; keep them the same.
- **authSecretFor()** in core.js must match the Android app's AuthRepository.kt, or app and web logins will disagree.
- **getRemarks()** exists in both core.js and admin-scoring.js; keep the bands identical.
- **Respect the load order** (section 10) and edit the last definition of an overridden function.
- Firestore may ask for a **composite index** for some queries ("failed-precondition"); the browser console prints a link that creates it.
- Campus networks: both apps enable *experimentalAutoDetectLongPolling* so Firestore works where streaming is blocked.
- Annex wording mirrors the QA office's signed copy and CMO 19; treat changes to annex text as changes to an official document.

Generated from the source code of the Faculty-Eval-System repository.